



DELHI PUBLIC SCHOOL NEWTOWN
SESSION: 2021-22
MONDAY TEST

CLASS: IX
SUBJECT: CHEMISTRY

FULL MARKS: 40
DATE: 19.7.2021

General Instructions:

- The paper consists of two printed pages.
- Answers should be to the point.
- Question numbers should be copied carefully while answering the questions.

Question:1

Name the following:

- i) The radical which is represented as MnO_4^{1-}
- ii) A chemical reaction in which an element replaces another element in a compound.
- iii) A metal which sinks in cold water turning the solution turbid.
- iv) A trivalent metal which displaces hydrogen from dilute sulphuric acid.
- v) The acid whose formula is $\text{H}_2\text{C}_2\text{O}_4$. [5]

Question:2

Give reasons for the following:

- i) A chemical equation should be balanced.
- ii) Lead cannot be used in the preparation of hydrogen using dilute sulphuric acid.
- iii) Hydrogen is not collected over air in the laboratory.
- iv) Mountaineers have to carry oxygen cylinders with them.
- v) Concentrated sulphuric acid is not used in the laboratory preparation of hydrogen gas. [5]

Question:3

What would you observe when :

- i) Potassium iodide solution is added to lead nitrate solution.
- ii) Steam is passed over aluminium.
- iii) Sulphur dioxide gas is passed through potassium dichromate solution.
- iv) Water is added to quicklime.
- v) Sodium hydroxide solution is added to copper sulphate solution. [5]

Question:4

Give balanced equation for the following word equations:

- i) Iron (III) oxide + sulphuric acid $\rightarrow$ Iron (III) sulphate + water.
- ii) Zinc + Sodium hydroxide $\rightarrow$ sodium zincate + hydrogen
- iii) Zinc sulphide + oxygen $\rightarrow$ zinc oxide + sulphur dioxide
- iv) Lead nitrate $\rightarrow$ lead monoxide + nitrogen dioxide + oxygen
- v) Ammonium dichromate $\rightarrow$ nitrogen + chromium oxide + water. [5]

Question:5

Write the chemical formula for the following compounds:

- i) Stannic phosphate
- ii) Calcium acetate
- iii) Potassium ferrocyanide
- iv) Sodium arsenate
- v) Aluminium borate

[5]

Question:6

The following questions relate to the laboratory preparation of hydrogen from zinc granules and a dilute acid.

- i) Granulated zinc is preferred to pure zinc for the reaction with dilute acid. Explain.
- ii) Can dilute nitric acid be used in the preparation of hydrogen? Justify.
- iii) How is the gas collected?
- iv) State one important precaution you should take during the preparation of the gas.
- v) How would you prove using a chemical test that the collected gas is hydrogen?[5]

Question:7

Solve the following numericals:

- i) One litre of a gas at 10°C is heated till both its volume and pressure are tripled. Find the new temperature.
- ii) What temperature would be required to reduce the volume of a gas to $\frac{1}{6}$ the initial volume at constant pressure if the gas was originally at STP conditions. State the Law.

[5]

Question:8

- i) Calculate the percentage composition of barium in barium nitrite. [Ba=137, N=14, O=16]
- ii) State the type of reaction in the following equations:
 - a) $2\text{Ca}(\text{NO}_3)_2 \rightarrow 2\text{CaO} + 4\text{NO}_2 + \text{O}_2$
 - b) $\text{Pb}(\text{OH})_2 + 2\text{HNO}_3 \rightarrow \text{Pb}(\text{NO}_3)_2 + 2\text{H}_2\text{O}$
 - c) $2\text{Al} + 3\text{CuSO}_4 \rightarrow \text{Al}_2(\text{SO}_4)_3 + 3\text{Cu}$.

[5]